SujetStore

Educational Archive Platform for the Algerian Curriculum

Technical Manual — Architecture, Features & Roadmap

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1. Introduction: Rethinking Digital Learning in Algeria

1.1. The Problem

In the current Algerian educational landscape, students face a fragmented digital experience. Exam papers, quizzes, and study materials are scattered across social media groups, unmoderated forums, and low-quality scan repositories. Key issues include:

- **Content Dispersion:** Thousands of documents are posted daily across Facebook groups and Telegram channels with no indexing, versioning, or quality control. Finding a specific exam from a specific trimester becomes an exercise in frustration.
- **No Pedagogical Structure:** Materials are dumped without classification by trimester, type, difficulty, or alignment with the official Ministry of Education curriculum.
- **Poor Visual Quality:** Most documents are low-resolution phone scans with inconsistent fonts, skewed pages, and illegible handwriting.
- **Hostile User Experience:** Existing platforms rely on aggressive advertising, intrusive pop-ups, and confusing navigation that actively hinders the learning process.

1.2. The Vision

SujetStore was conceived as a purpose-built, open-source educational archive platform that directly addresses these pain points. It provides a clean, focused, and fast experience for browsing, filtering, and studying exam papers, interactive quizzes, and study materials for the Algerian curriculum — from primary school through high school.

1.3. Core Objectives

- **Absolute Focus:** Deliver content within a distraction-free, ad-free interface designed for deep study and review. No pop-ups, no banners, no autoplay videos.
- **Strict Taxonomy & Exam Codes:** Organize the entire curriculum using the official Algerian Exam Code system, ensuring accurate subject mapping across 9 different High School streams (e.g., distinguishing Physics for Math vs. Science streams) avoiding overlap.
- **Hierarchical Archiving:** A clear, navigable taxonomy: *Level* $\longrightarrow$ *Year* $\longrightarrow$ *Stream (Secondary)* $\longrightarrow$ *Subject* $\longrightarrow$ *Trimester* $\longrightarrow$ *Document Type.*
- **Interactive Learning:** Move beyond static PDFs by providing a robust 14-type quiz engine that tests comprehension instantly.
- **Empowering Content Creators:** Provide a powerful yet simple admin CMS that allows educators to upload, categorize, and manage documents and interactive quizzes effortlessly.

- **Resilient Infrastructure:** Leverage a modern, lightweight stack (SvelteKit + Bun + SQLite) that can scale to millions of concurrent users without infrastructure collapse.

2. System Architecture & Tech Stack

SujetStore is built on a modern, performance-oriented stack carefully selected for speed, simplicity, and developer experience.

2.1. The Tech Stack

Component	Technology	Rationale
Runtime	Bun (v1.1+)	3× faster than Node.js. Built-in bundler, test runner, and native SQLite driver.
Framework	SvelteKit (Svelte 5)	Full-stack framework with SSR. Powerful reactive runes (`$state`, `$derived`). Minimal client-side overhead.
Database	SQLite	Embedded, zero-config, read-optimized. Uses `bun:sqlite` for native performance.
ORM	Drizzle ORM	Lightweight, type-safe query builder with full TypeScript support.
CSS	TailwindCSS v4	Utility-first CSS with native CSS variable support and zero-config setup.
UI Components	Shadcn-Svelte	Accessible, themeable component library with semantic color tokens.
Auth	Custom (bcryptjs)	Session-based authentication stored in SQLite. No external dependencies.
Deployment	Caddy + systemd	Auto-SSL reverse proxy with process management.

2.2. Architecture Philosophy: The Compact Monolith

Rather than adopting microservices or a headless frontend/backend split, SujetStore uses a **Compact Monolith** architecture:

- **Single Process:** The frontend (Svelte 5) and backend (SvelteKit server routes) run within a single Bun process.
- **File-Based Routing:** Routes are defined by the filesystem structure under `src/routes/`, eliminating the need for manual route configuration.
- **Co-located Data:** Both SQLite databases (`content.db` and `users.db`) live on the same server, accessed via `bun:sqlite` for sub-millisecond query performance.

2.3. Dual-Database Strategy

The system uses two separate SQLite databases to cleanly separate concerns:

2.3.1. Content Database (`content.db`) — Read-Intensive

Stores all educational content: levels, years, subjects, trimesters, documents, quizzes, and quiz questions. This database is generated offline using the seed script and deployed as a single file. Updates are often performed by replacing the entire file or making bulk inserts through the Admin panel.

2.3.2. Users Database (`users.db`) — Read-Write

Stores admin user accounts, session data, user gamification data (points, badges), quiz attempts, ratings, comments, and global notifications. WAL mode is enabled for concurrent read/write access:

```
PRAGMA journal_mode = WAL;
PRAGMA busy_timeout = 5000;
PRAGMA synchronous = NORMAL;
```

This ensures that high-volume writes (like tracking points and attempts) never block reads on the content database.

2.4. The Exam Paper Coding System

A core feature of SujetStore is its adherence to the **Algerian Exam Coding System**. This handles the complex relationship between educational streams (شعب) and subject papers.

2.4.1. The Golden Rule

The system follows a strict principle: *"If two streams take the exact same exam paper, they share a single Exam Code."*

- **Shared Papers:** If "Physics" for the *Experimental Sciences* stream is identical to "Physics" for the *Math* stream, both map to a single `PHY-SCI` code.
- **Unique Papers:** If "Math" for *Math* stream is different from "Math" for *Literary* stream, they receive separate `MAT-MTM` and `MAT-LIT` codes respectively.

2.4.2. Relational Mapping

The SQLite schema implements this through three dedicated tables:

- `streams`: Defines secondary education tracks (e.g., GEN, SE, MATH).
- `level_streams`: Maps specific years (e.g., 2AS, 3AS) to available streams.
- `stream_subjects`: Maps streams to their respective exam-code subjects and trimesters, ensuring students only see relevant materials for their track.

2.5. Development Workflow

```bun
# Install dependencies using Bun
bun install

# Seed the databases with the Algerian default tree (Primary, Middle, Secondary)
bun run scripts/seed-education.ts
bun run scripts/seed-algerian-questions.ts

# Start the development server
bun run dev

# Type-check the project
bunx svelte-check
```

3. UI/UX & Design System

SujetStore was designed with the principle that *"simplicity creates effectiveness."* The interface feels native, fast, and accessible regardless of the device.

3.1. Native RTL & Arabic Typography

The entire design is built from the ground up for Right-to-Left (RTL) text direction.

- **Typography:** The `Cairo` font from Google Fonts is applied globally. It was selected for its exceptional legibility in Arabic numerals and varied stroke weights, making it ideal for both long reading (like document summaries) and bold headers.
- **Layout Engine:** Flexbox and CSS Grid are utilized exclusively with logical CSS properties (`padding-inline`, `margin-inline-start`) to ensure perfect mirroring natively without any hacky overrides.

3.2. Dark & Light Mode with Shadcn-Svelte Theming

The platform implements a complete dual-theme system using `Shadcn-Svelte` semantic color tokens. Every component — from buttons and dropdowns to cards and toggle switches — automatically adapts to the user's preferred color scheme based on their system setting, with a manual override toggle available in the navigation bar.

Key semantic tokens used throughout the interface:

- `bg-background` / `text-foreground` — Base page colors
- `bg-card` / `text-card-foreground` — Elevated surface containers (like Document boxes)
- `bg-muted` / `text-muted-foreground` — Subdued backgrounds and secondary helper text

- `bg-primary` / `text-primary-foreground` — Active elements, CTAs, and primary buttons
- `bg-accent` / `text-accent-foreground` — Hover and interactive states

This approach guarantees that **no hardcoded hex colors** appear in the core UI codebase (excluding specific gamification badges), ensuring perfect visual contrast in both modes.

3.3. Mobile-First Responsive Design

Recognizing that the vast majority of Algerian students access educational content from smartphones, the interface implements strict mobile-first paradigms:

- **Stacked Controls:** Filter controls stack vertically on small screens and align horizontally on desktop displays.
- **Horizontal Scrolling:** Tab navigation (for trimesters and quiz stats) scrolls horizontally with hidden scrollbars when screen width is constrained to preserve vertical screen real estate.
- **Touch Targets:** All interactive elements meet or exceed the minimum touch target size of 44×44px.
- **Glassmorphism Architecture:** The UI utilizes a premium "Glassmorphism" effect for year and subject cards. This involves backdrop-blur filters combined with semi-transparent borders and subtle gradients, creating a multi-layered depth effect that remains performant on mobile devices.
- **Data Visualization:** Cards dynamically display item counts (e.g., "15 Subjects", "24 Exams"), providing students with immediate visual cues about content density before navigation.

3.4. Global Navigation Hierarchy

A global breadcrumb navigation system (`Level / Year / Subject`) is present on all inner pages. This ensures the student always knows their exact location in the curriculum taxonomy and can navigate back to a parent category with a single click, completely eliminating the feeling of being "lost" inside the application.

4. The Document Engine

The core utility of SujetStore is the rapid discovery, filtering, and consumption of static educational documents (Exams, Assignments, Summaries).

4.1. Discovery & Categorization

The homepage presents the three educational levels of the Algerian system:

- **Primary** — Grades 1 through 5
- **Middle School** — Grades 1AM through 4AM
- **Secondary** — Grades 1AS through 3AS (across all specialized streams)

Each drill-down reveals a structured grid of years, leading to the core subjects.

4.2. Trimester Tabs: Segmented Control

Inside a subject page, documents are grouped using a **Segmented Control** interface — a horizontal tab bar with three segments: *Term 1*, *Term 2*, *Term 3*, and *General*.

Built on Shadcn UI's `Tabs` pattern, this control physically separates the academic year. Furthermore, the system implements **Smart Auto-Tab Selection**: On page mount, the engine scans the trimesters in order and automatically activates the first one that contains available documents, saving the user from clicking through empty tabs.

4.3. The Contextual Filter Bar

Inside each trimester tab, users can narrow down hundreds of documents instantly using client-side Svelte 5 `$derived` runes:

- **Document Type:** Filter by Exam, Assignment, Lesson, Summary, or Book.
- **Source:** Filter by Official (Ministry standard) or Proposed (Teacher drafts).
- **Strict Solution Toggle:** A prominent switch to show "Only documents with solutions attached."

Because filtering happens entirely client-side, the UI updates instantly without any network latency or skeleton loaders.

4.4. Smart PDF Viewer & Printing

SujetStore integrates an in-browser PDF viewing experience that keeps students on the platform without forcing downloads:

- **Fullscreen Modal:** Clicking a document opens a distraction-free, full-height modal containing the PDF `iframe`.
- **One-Click Solution Toggle:** If a document possesses an attached solution file, a floating toggle button allows the student to flip instantly between viewing the "Subject" and the "Correction" without reloading the modal.
- **Background Printing:** Print functionality is handled seamlessly, opening the system print dialog directly from the file blob to ensure perfect margins regardless of browser quirks.

4.5. Smart Tracking Persistence

To help students track what they have studied over long periods, the application implements local persistence tracking. When a student opens a document, its ID is saved to the browser's `localStorage` via a `Set` implementation.

When displaying the document lists, previously viewed items are marked with a distinct Checkmark indicator (✓) and a slightly faded styling, providing immediate visual feedback on their revision progress.

5. The Interactive Assessment Engine

SujetStore features an incredibly robust, deeply integrated interactive quiz engine that goes far beyond simple multiple-choice questions. It provides a full-featured Moodle-like assessment environment tailored specifically for the Algerian curriculum, complete with 14 distinct question types, visual interactive editors, and a centralized question bank.

5.1. 1. Admin Quiz Management System

The Quiz Management System is designed to give administrators and educators total control over interactive content.

5.1.1. 1.1. Quiz Creation and Configuration

Administrators can create quizzes and bind them to specific `year_subjects` (e.g., "Math - 2nd Year Secondary - Science Stream") and, optionally, specific trimesters. Each quiz can be configured with:

- **Difficulty Levels**: Easy, Medium, Hard (affecting gamification point yields).
- **Monetization/Premium Gating**: Quizzes can be marked as `isPremium`. Free users cannot access premium quizzes, driving subscription conversions.
- **Publishing States**: Quizzes can be kept as drafts (`isPublished = false`) while under construction.
- **Time Limits & Passing Scores**: Formal assessment parameters to simulate real exam conditions.

5.1.2. 1.2. Quiz Duplication

To maximize educator productivity, the system features a one-click **Quiz Duplication** mechanism. Clicking "Copy" on a quiz triggers a backend API (`/api/admin/quizzes/[id]/duplicate`) that:

- Deep-clones the quiz metadata.
- Appends "(Copy)" to the title.

- Resets the `isPublished` flag to `false`.
- Duplicates all relation links (`quiz_questions`) while preserving the exact ordering and point values of the original quiz, without duplicating the underlying questions in the bank.

5.2. 2. The Centralized Question Bank

Instead of siloing questions within individual quizzes, SujetStore utilizes a **Centralized Question Bank**. All questions are stored in a single table (`questions`) and mapped to quizzes via a many-to-many relationship table (`quiz_questions`).

5.2.1. 2.1. Benefits of the Question Bank

- **Reusability**: A single "Algebra Hard Question" can be linked to a Midterm Quiz, a Final Exam Quiz, and a Premium Practice Quiz simultaneously.
- **Data Integrity**: Updating a question's typo in the bank instantly fixes it across all quizzes that use it.
- **Categorization**: Questions can be organized into `question_categories` (e.g., "Grammar", "Mechanics", "Fractions"), allowing for structured storage.

5.2.2. 2.2. Visual Identification

The Question Bank UI is designed for rapid parsing. Each of the 14 question types is represented by a unique, color-coded Lucide icon (e.g., a green check-square for True/False, a purple mapping pin for Hotspots). This visual taxonomy persists across the bank, the builder, and the preview modals.

5.3. 3. The Quiz Builder Interface

The Quiz Builder is a highly interactive, drag-and-drop SPA (Single Page Application) built with Svelte 5.

5.3.1. 3.1. Drag-and-Drop Sequencing

Educators can visually reorder questions within a quiz using HTML5 Drag-and-Drop API. The new order is automatically saved and synchronized with the backend.

5.3.2. 3.2. JSON Import Engine

For massive bulk operations, the builder includes a **JSON Import Engine**. Educators can author hundreds of questions in JSON format (usually generated via AI or exported from other systems) and ingest them instantly. The specific API parses the JSON array, validates the schema for the requested question types, inserts them into the Question Bank, and automatically links them to the active quiz.

5.3.3. 3.3. Question Bank Modal

Educators can click "Import from Bank" to open a modal that searches the entire database. They can filter by Type, Difficulty, and Search Query, then multi-select questions to inject into the current quiz.

5.3.4. 3.4. Live Previewing

Every question can be previewed directly from the Bank or the Builder. The preview modal renders the exact Svelte component that the student will see, and includes a test validation engine (`checkPreviewAnswer()`) to ensure the grading logic works perfectly before publishing.

5.4. 4. The 14 Question Types in Detail

SujetStore supports an unparalleled variety of 14 interactive question types, all serialized seamlessly into a single `question_data` JSONB column.

5.4.1. 4.1. Standard & Text Types

1. **Multiple Choice (MCQ)**: Supports single or multiple correct answers. Arabic lettering (أ، ب، ج، د) is automatically applied for localization. Options are randomized at runtime.
2. **True/False**: A streamlined binary choice with bold, recognizable UI cards.
3. **Short Answer**: Free-form text input. Evaluated server-side by checking if the student's string contains a minimum number of required predefined `keywords` (allowing for typographical leniency).
4. **Fill in the Blank**: A paragraph with blank text inputs. Evaluated by strict or case-insensitive string matching array.
5. **Cloze (Dropdowns)**: Similar to Fill-in-the-Blank, but replaces free-text inputs with HTML `<select>` dropdowns defined inline.
6. **Essay/Long Answer**: A rich text or plain text area that tracks word counts. Graded manually, or automatically via keyword density analysis.

5.4.2. 4.2. Complex Interactive Types

7. **Matching**: Students draw connections or select corresponding pairs from two columns (e.g., matching a philosopher to their quote).
8. **Ordering**: A vertical stack of draggable items (e.g., historical events). Students drag them to arrange them in chronological order.
9. **Drag & Drop (Categorization)**: Students drag draggable pills into designated drop-zone buckets.

10. **Sentence Reorder**: Designed for language learning. A scrambled list of words that the student must tap/drag sequentially to form a syntactically correct sentence.
11. **Matrix**: A 2D grid where rows are statements and columns are evaluation criteria (e.g., mapping 5 different animals to "Mammal", "Reptile", or "Bird" via radio buttons).

5.4.3. 4.3. Advanced Mathematical Types

12. **Calculated**: A highly dynamic mathematical engine. The teacher defines a formula (e.g., `{x} * {y} + 10`). At runtime, the server generates random values for `x` and `y` within defined minimum/maximum boundaries. The student is presented with the unique numbers, and their text input is evaluated against the dynamically computed mathematical result, incorporating an acceptable `tolerance` margin for floating-point errors.

5.4.4. 4.4. Visual & Image Editors (The Image Upload API)

To support visual questions, SujetStore includes an internal `/api/admin/upload-image` endpoint that handles secure file uploads (WebP, PNG, JPEG), returns a public path, and integrates directly into the visual editors.

13. **Hotspot**: The teacher uploads an image (e.g., a map of Algeria or a human heart). The admin UI provides a **Visual Click-to-Place Editor**. The teacher clicks directly on the image to define circular "Zones". They then designate one or more zones as the correct answer. The student sees the image and must click the correct physical coordinate.
14. **Drag to Image**: The teacher uploads a background image and defines text "Labels". Using the Visual Editor, the teacher drags the labels to specific coordinates on the image to define their `correctX` and `correctY` positions. The student receives the image with labels clustered at the bottom, and must drag them to their correct physical overlays within a 10px tolerance threshold.

5.5. 5. Grading and Security Architecture

All evaluation logic runs **Server-Side**. When a student clicks "Check Answer" or "Submit Exam":

1. Only the student's payload (e.g., `selectedZone: 2` or `order: [3, 1, 2]`) is dispatched to the server.
2. The server loads the true schema from the database (which was never sent to the client, preventing DevTools cheating).
3. The server runs the specific evaluator for the question type.

4. The server responds with `isCorrect: true/false`, the points earned, and optionally the `correctAnswer` payload so the UI can highlight mistakes in red and correct answers in green.

6. Gamification & Engagement

To increase student engagement, motivation, and retention, SujetStore implements a complete gamification layer. This transforms the solitary act of studying into a rewarding, trackable journey.

6.1. Points System

The foundation of the gamification system is the Points ledger. Points are awarded automatically for positive educational behaviors:

- **Completing a Quiz:** +10 points.
- **Achieving a Perfect Score:** Bonus points depending on quiz length.
- **Earning a Badge:** Bonus points ranging from +10 to +50 based on difficulty.

All point transactions are recorded in the `user_points` table with a timestamp and reason, providing students with a transparent history of their earnings.

6.2. Achievement Badges

The platform defines 8 built-in achievement badges that students unlock by reaching specific milestones in the Quiz Engine.

Badge	Icon	Unlock Condition
First Quiz	🎯	Complete your very first interactive quiz
Quiz Explorer	📚	Complete 5 total quizzes
Quiz Master	🏆	Complete 10 total quizzes
Perfect Score	⭐	Score 100% on any quiz
Speed Demon	⚡	Complete a quiz in under 1 minute
Hot Streak	🔥	Score 80%+ on 3 consecutive quizzes
Versatile	🎪	Try 5 different interactive question types
Centurion	💯	Accumulate 100 total points

The badge evaluation logic runs asynchronously upon quiz submission. When a new threshold is met, the system inserts a record into the `user_badges` table.

6.3. The Achievements Dashboard

Students can view their progress on the dedicated Achievements page (`/achievements`). This page utilizes a horizontal tab layout to present three key views:

1. **Badges Grid:** A visual display of all 8 badges. Unearned badges appear grayscale/locked, while earned badges are fully colored with their unlock timestamps.
2. **Leaderboard:** A global ranking showing the top 10 most active students on the platform, ranked by total accumulated points.
3. **Points History:** A chronological log of the student's recent point earnings and the specific actions that triggered them.

7. Administrative Dashboard & Content Management

A platform is only as good as its content. SujetStore provides administrators and educators with a powerful, custom-built Content Management System (CMS) to manage the entire educational tree without touching the database.

7.1. Security & Authentication

Access to the `/admin` routes is strictly protected.

- Authentication uses a custom implementation relying on `bcryptjs` for password hashing.
- Sessions are stored in the `users.db` SQLite database via secure HTTP-only, `SameSite=Lax` cookies.
- Rate limiting is enforced on login endpoints to prevent brute-force attacks.

7.2. The Document CMS

The administration panel provides full CRUD (Create, Read, Update, Delete) capabilities for the taxonomic hierarchy:

1. **Levels** (e.g., Middle School)
2. **Years** (e.g., 4AM)
3. **Subjects** (e.g., Mathematics)
4. **Documents** (e.g., Term 1 Exam with Solution)

Key UI Features:

- **Modal-Based Editing:** All add/edit operations use floating Shadcn-Svelte modals for fast interaction without triggering full page reloads.
- **Confirmation Dialogs:** Destructive operations require explicit confirmation through a `ConfirmModal` component to prevent accidental data loss.

- **Secondary Stream Awareness:** The CMS is fully aware of the Algerian stream system. When uploading documents or quizzes for secondary school levels, the system prompts for stream scoping (e.g., scoping a Physics exam specifically to the *Mathematical Technical* track) to ensure accurate content delivery without subject name collisions.

7.3. The Interactive Quiz Builder

Creating interactive Moodle-style content is notoriously complex in most LMS platforms. SujetStore solves this by providing a highly optimized, visual Quiz Builder.

Builder Features:

- **Drag-and-Drop Ordering:** Teachers can reorder questions within a quiz visually.
- **Specialized Question Sub-forms:** Because the platform supports 14 different question types, the builder dynamically renders a tailored sub-form based on the selected type.
  - For example, selecting "Matrix" reveals a grid builder for rows and columns.
  - Selecting "Hotspot" reveals an image upload field and X/Y coordinate inputs for defining clickable zones.
- **Quiz Metadata Engine:** Set time limits, required passing scores, and publish toggles.

7.4. Analytics Dashboard

The admin home page provides a high-level overview of platform health:

- Total Document counts grouped by Level and Year.
- Global point accumulations.
- Quiz completion statistics to identify the most difficult or popular exercises across the platform.

8. Deployment & Future Roadmap

8.1. Production Deployment Guide

SujetStore is designed to be easily deployed on cheap VPS infrastructure (e.g., Contabo, Hetzner) while handling massive scale.

8.1.1. Current Infrastructure Footprint

Component	Specification
Provider	Standard Linux VPS (Ubuntu 22.04+)
Web Server	Caddy (auto-SSL reverse proxy)
Process Manager	systemd service unit
Backup Strategy	Daily SQLite `.backup` of `users.db` with 7-day retention

8.1.2. Deployment Commands

A deployment script (`deploy.sh`) is provided in the repository root to automate the build and restart process.

```
./deploy.sh full    # Initial setup, dependency install + deployment
./deploy.sh deploy  # Subsequent updates (git pull, build, restart)
```

8.2. Security Layers

Layer	Implementation
Transport	HTTPS enforced via Caddy auto-SSL
Cookies	`HttpOnly` · `Secure` · `SameSite=Lax`
Passwords	bcryptjs hashing for Admin accounts
SQL Injection	Guaranteed prevention via Drizzle ORM prepared statements

8.3. Future Roadmap

While Phase 1 (The Interactive Assessment Engine) is now successfully completed, SujetStore has clear expansion vectors for Phase 2 and beyond.

8.3.1. Phase 2: Video Solutions & Streaming Architecture

To support deeper understanding—especially for subjects like Physics and Mathematics—the platform will integrate video walkthroughs mimicking a teacher's explanation.

Proposed Streaming Tech Stack:

- **HLS Protocol:** Adaptive bitrate streaming with 10-second chunks for instant playback.
- **AES-128 Encryption:** Per-chunk encryption to prevent unauthorized downloads.
- **Backblaze B2:** Cold storage at a fraction of the cost of Amazon S3.
- **Cloudflare CDN:** Zero egress fees via the Bandwidth Alliance, providing rapid edge distribution across North Africa.

8.3.2. Phase 3: Advanced Analytics & Weakness Detection

Transitioning from macro-level admin analytics to personalized student diagnostics:

- **Automated Weakness Detection:** Analysis of specific interactive quiz errors to identify targeted knowledge gaps (e.g., "Struggles with Fractions").
- **Personalized Recommendations:** Automatically suggesting specific summaries or lessons based on failed quiz competencies.
- **Parent Dashboards:** Summary statistics tracking a student's study time, exam scores, and progression by trimester.

9. The Educational Hierarchy Engine

SujetStore models the complex, multi-tiered Algerian educational system with high precision. Understanding this hierarchy is crucial for administrators managing documents and quizzes, as it dictates how content is routed to specific students.

9.1. 1. The Core Entities

Our database schema breaks the educational journey into four distinct mathematical entities: **Levels**, **Years**, **Streams**, and **Subjects**.

9.1.1. 1.1. Levels (المراحل التعليمية)

Levels represent the broadest categorization in the Algerian system (`education_levels` table):

- **Primary Education (التعليم الابتدائي)**: 5 years of schooling.
- **Middle Education (التعليم المتوسط)**: 4 years of schooling culminating in the BEM exam.
- **Secondary Education (التعليم الثانوي)**: 3 years of schooling culminating in the BAC exam.

9.1.2. 1.2. Years (السنوات الدراسية)

Each Level is subdivided into specific Years (`years` table).

- For Primary: 1AP, 2AP, 3AP, 4AP, 5AP.
- For Middle: 1AM, 2AM, 3AM, 4AM.
- For Secondary: 1AS, 2AS, 3AS.

Years are strictly parented to a single Level (`level_id` foreign key).

9.1.3. 1.3. Streams (الشعب الدراسية)

Streams represent the specialized academic tracks that students follow. In the Algerian system, specialization begins in Secondary Education (`streams` table). Examples of Streams include:

- **Common Trunk Science & Tech (جذع مشترك علوم وتكنولوجيا)** - Typically 1AS.
- **Common Trunk Letters (جذع مشترك آداب)** - Typically 1AS.
- **Experimental Sciences (علوم تجريبية)** - 2AS and 3AS.
- **Mathematics (رياضيات)** - 2AS and 3AS.
- **Management and Economics (تسيير واقتصاد)** - 2AS and 3AS.
- **Foreign Languages (لغات أجنبية)** - 2AS and 3AS.

To map which streams are available in which years, the database uses a many-to-many join table (`level_streams`). For example, the **Experimental Sciences** stream is mapped to **2AS** and **3AS**, but never **1AS**, **4AM**, or **1AP**. Primary and Middle education act functionally as a

single “General” stream per year, although the architecture supports adding arbitrary streams if the educational system changes.

9.1.4. 1.4. Base Subjects (المواد الأساسية)

The `subjects` table holds the normalized names of subjects (e.g., “Mathematics”, “Physics”, “Arabic Language”). These are abstract entities. They do not belong to a year or a stream; they are just labels and icons.

9.2. 2. The Intersection: YearSubjects and StreamSubjects

To attach actual educational content (PDF documents, interactive quizzes) to the hierarchy, we must define exactly **when** and **where** a base subject is taught.

9.2.1. 2.1. Year Subjects (مواد السنة)

The `year_subjects` table connects a Base Subject to a specific Year.

- **Examples**: “Math for 1AM”, “Physics for 3AS”.

When an admin navigates the backend dashboard (e.g., clicking on Secondary -> 3AS -> Physics), they are viewing a `year_subject`. All documents and quizzes are ultimately parented to a `year_subject_id`.

9.2.2. 2.2. The Complexity of Secondary Education (Shared vs. Specialized)

While `year_subjects` map subjects to years, Secondary Education introduces a severe complication: **Streams**. In 3AS (3rd Year Secondary), Physics is taught to both the **Experimental Sciences** stream and the **Mathematics** stream.

Must the admin upload the “2024 Final Physics Exam” twice—once for Experimental Sciences and once for Mathematics? **No.** SujetStore implements the **Algierian Exam Paper Coding System** via the `stream_subjects` table.

9.2.3. 2.3. Stream Subjects and the Exam Code Rule

The `stream_subjects` join table maps a `year_subject` to a combination of `stream` and `trimester`. The Golden Rule of SujetStore is: *“If two streams take the exact same physical piece of paper in an exam, they share the exact same exam code and thus the same `year_subject`.”*

Scenario A: Shared Subjects (المواد المشتركة) Consider the Islamic Sciences exam in the BAC. Every single stream (Math, Sciences, Letters, Management, etc.) takes the exact same exam paper on the same day.

- The admin uploads the “2024 Islamic Sciences BAC” document **once** to the `year_subject` “Islamic Sciences - 3AS”.
- In the `stream_subjects` table, this single `year_subject_id` is linked to **all six** 3AS streams.

- If a Math student or a Letters student browses their unique stream profile, they will both see this shared document. If the admin fixes a typo in the title, it updates instantly for all streams.

Scenario B: Specialized Subjects (المواد الخاصة) Consider Physics in 3AS. The **Mathematics** stream takes a much harder Physics exam than the **Experimental Sciences** stream. They take different papers on different days.

- To handle this, the system creates **two distinct** `year_subjects`:
  1. "Physics (Math Stream) - 3AS"
  2. "Physics (Science Stream) - 3AS"
- The admin uploads the Math Physics exam to year_subject 1, and the Science Physics exam to year_subject 2
- In `stream_subjects`, year_subject 1 is mapped **only** to the Math stream, and year_subject 2 is mapped **only** to the Experimental Sciences stream.

9.3. 3. Dynamic Trimesters

The Algerian school year is divided into three trimesters (الفصل الأول، الثاني، الثالث), plus General (عام) files for things like full-year curriculums or BAC subjects.

- Documents are assigned an optional `trimester_id`.
- Quizzes are assigned an optional `trimester_id`.
- The user interface automatically clusters and tabs content horizontally based on their trimester, calculating the exact document count for each trimester on the fly ensuring empty trimesters are hidden.